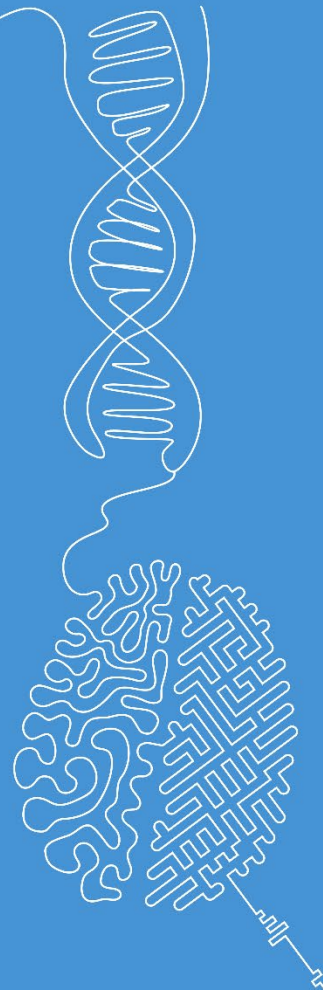


# Computed Tomography

Imaging for the Life Sciences

Robert Vorburger



# Content

- History
- Principles
- Applications
- Data Aspects



# Content

- **History**
- Principles
- Applications
- Data Aspects

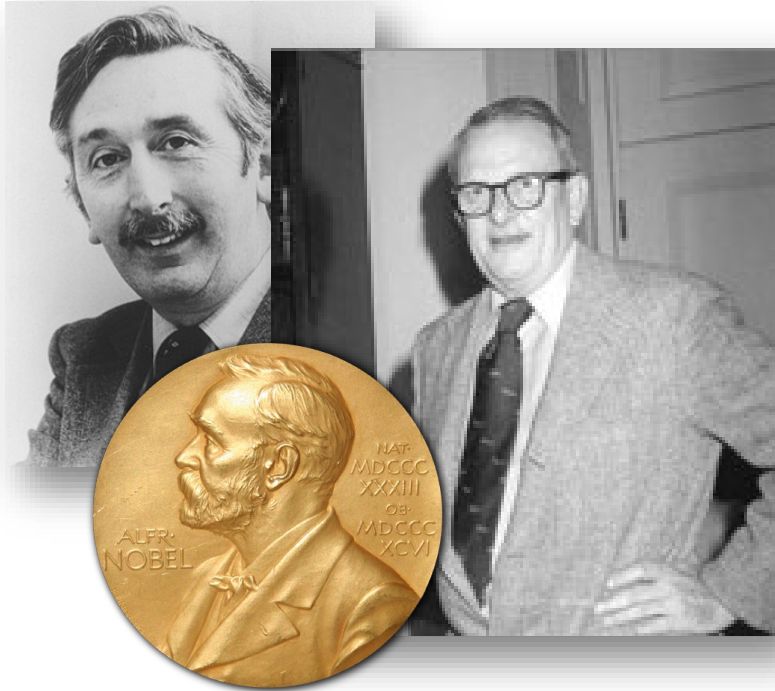


# Sir Godfrey Hounsfield



first commercially viable CT scanner was invented by  
Godfrey Hounsfield at EMI Ltd  
(1972)

# Allan MacLeod Cormack

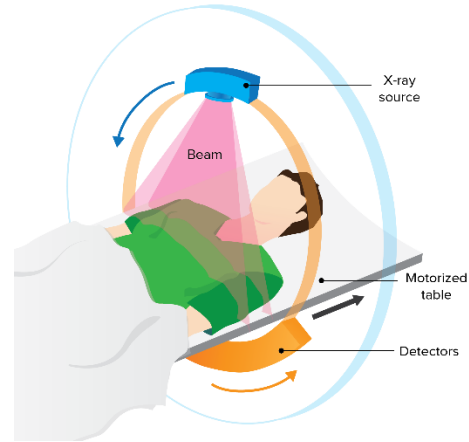


## **Nobel prize in medicine 1979**

Awarded to both Hounsfield and Cormack, which both independently worked on CT. Cormack provided theoretical insights in the 1960s, while Hounsfield built the first CT scanner in the 1970s

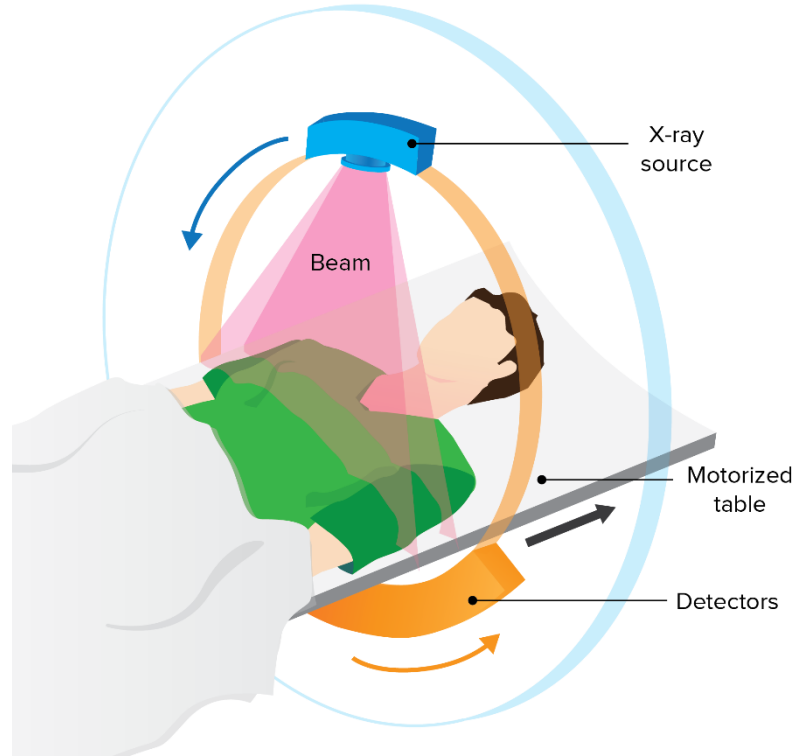
# Content

- History
- **Principles**
- Applications
- Data Aspects



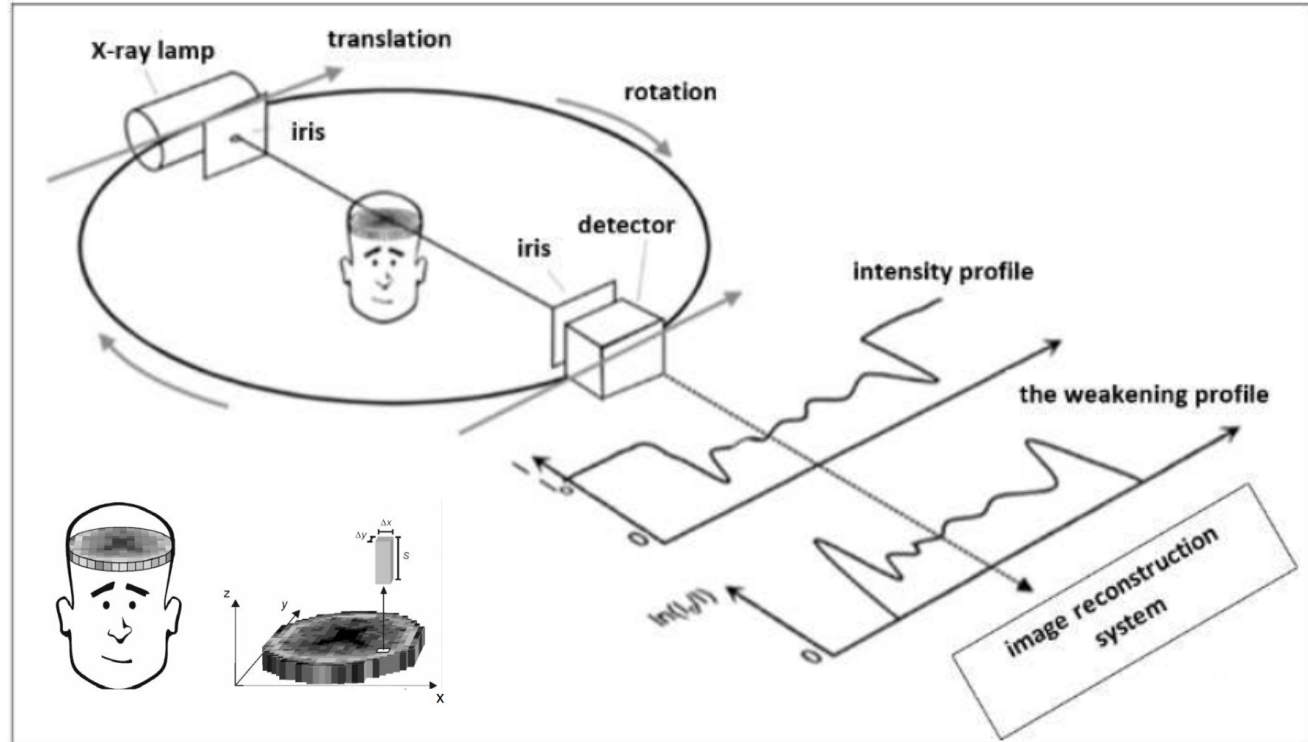
# Tomographic imaging

## Basic Setup



# Tomographic imaging

## Basic Setup





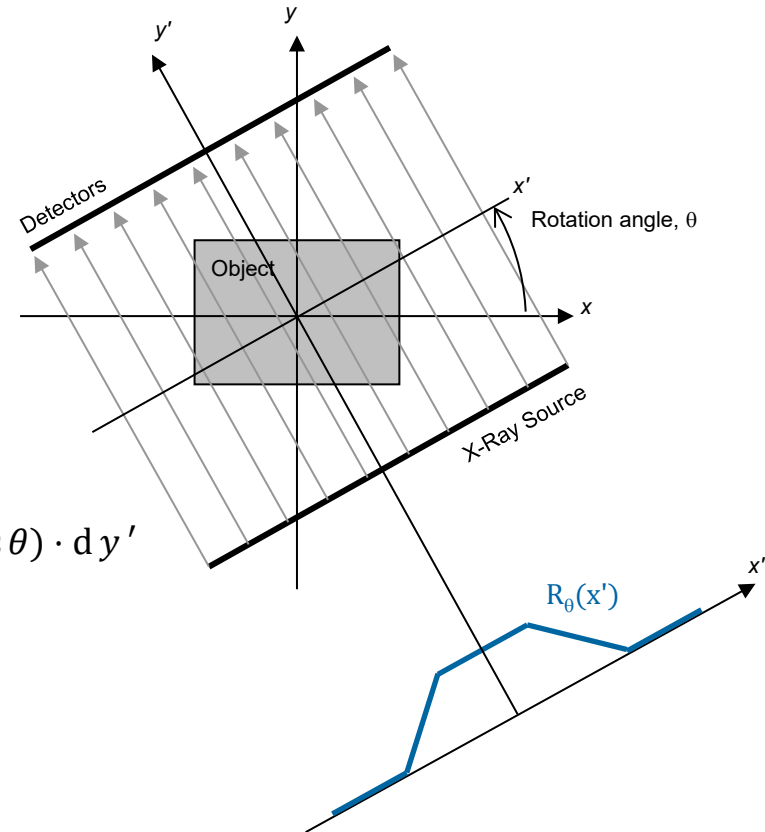
# Radon-Transformation

A 2D object can be defined by a two-dimensional function  $f(x,y)$ . The Radon transform  $R$  of  $f(x,y)$  is the line integral of  $f$  along the  $y'$ -axis.

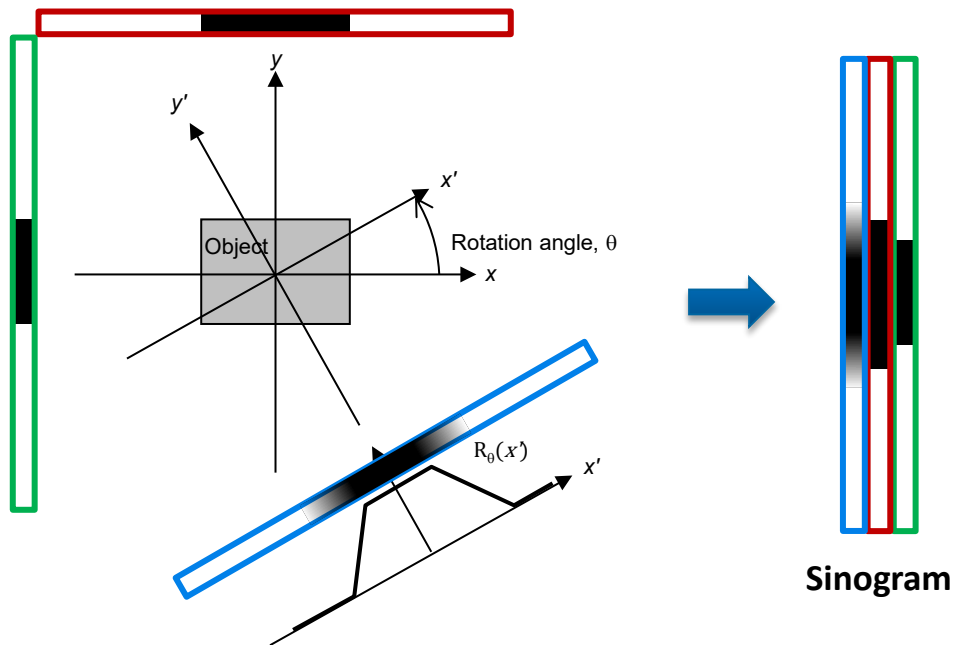
$$R_{\theta}(x') = \int_{-\infty}^{+\infty} f(x' \cdot \cos \theta - y' \cdot \sin \theta, x' \cdot \sin \theta + y' \cdot \cos \theta) \cdot dy'$$

where  $x'$  and  $y'$  represent the rotated axes:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$



# Sinogram

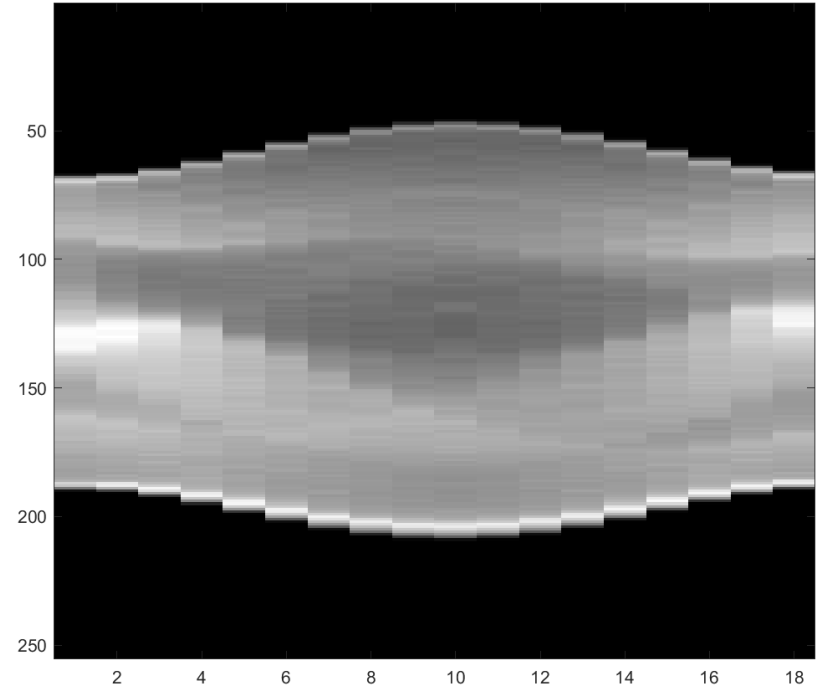


# Sinogram

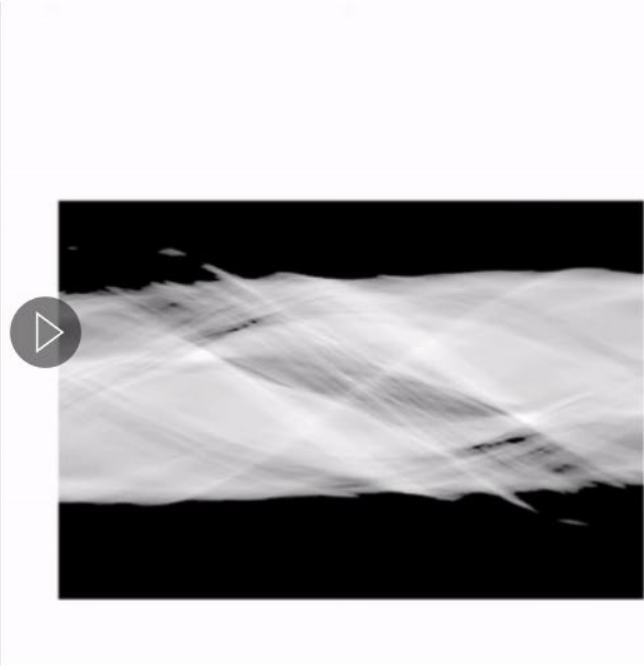
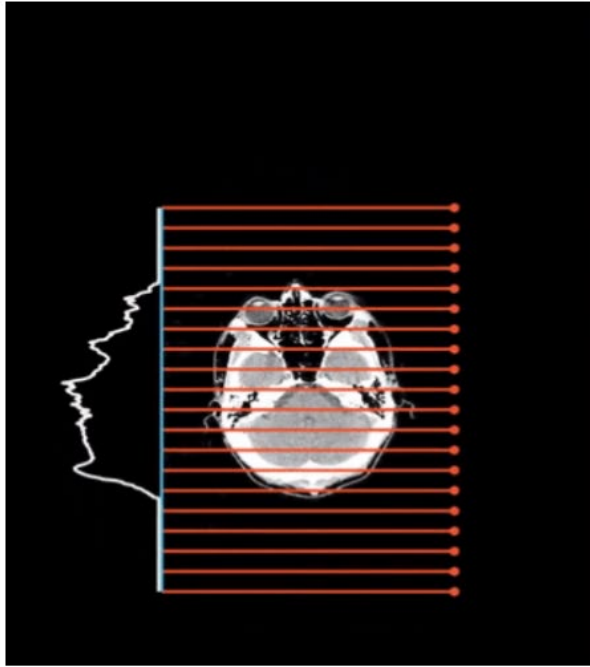
In CT imaging, a sinogram is a matrix in which each column corresponds to a Radon transformation  $R_\theta$ .

The number of columns is therefore given by the number of applied rotation angles  $\theta$ .

The number of rows corresponds to the resolution of the CT detector.



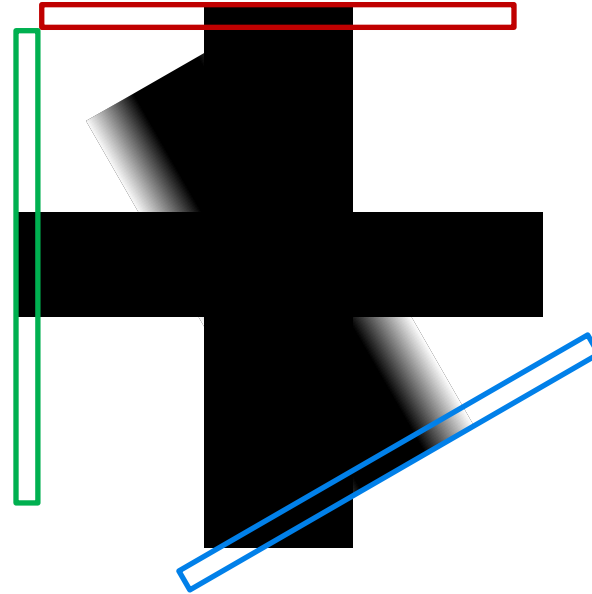
# Sinogram



Samuli Siltanem's side channel: [https://www.youtube.com/watch?v=q7Rt\\_OY\\_7tU](https://www.youtube.com/watch?v=q7Rt_OY_7tU)

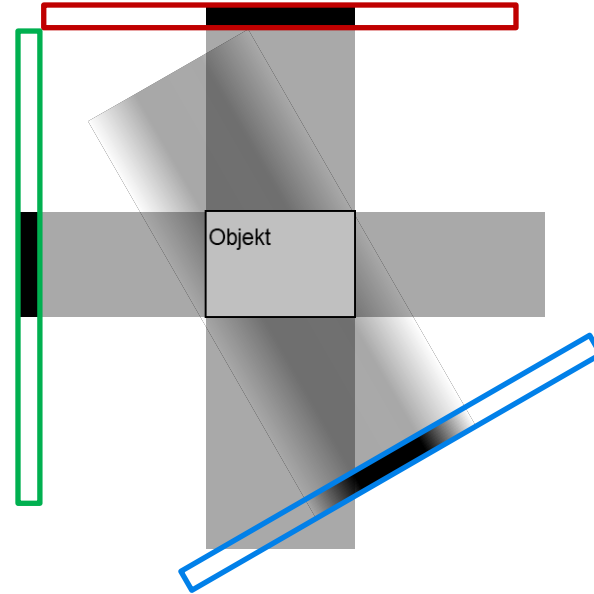
# Tomographic Imaging

1. Iteratively "smear back" projection rows (column vectors of the sinogram) with the same angle  $\theta$  as during recording and add up.
2. At the end, normalize and filter the reconstructed image.

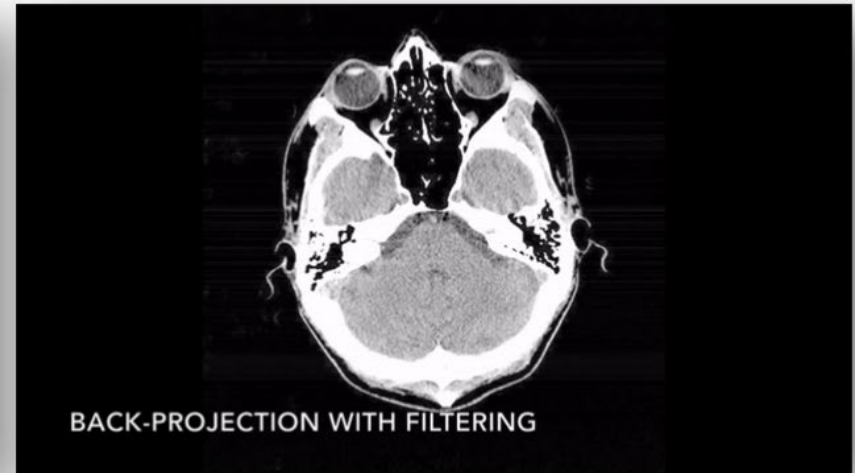
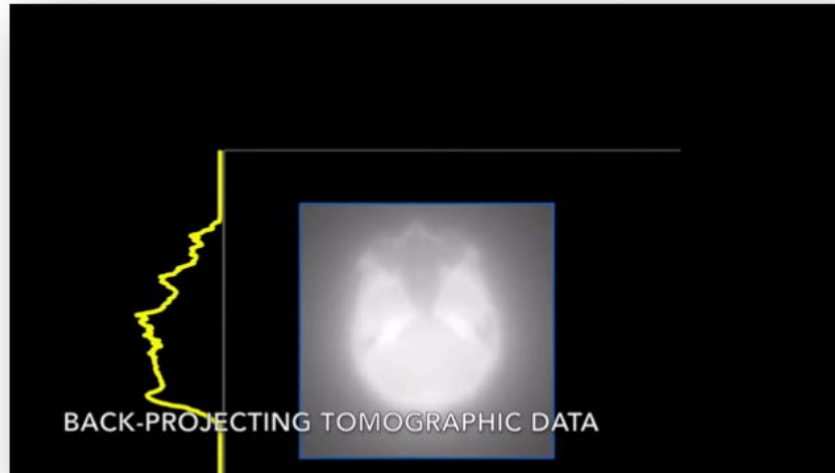


# Tomographic Imaging

1. Iteratively "smear back" projection rows (column vectors of the sinogram) with the same angle  $\theta$  as during recording and add up.
2. At the end, normalize and filter the reconstructed image.



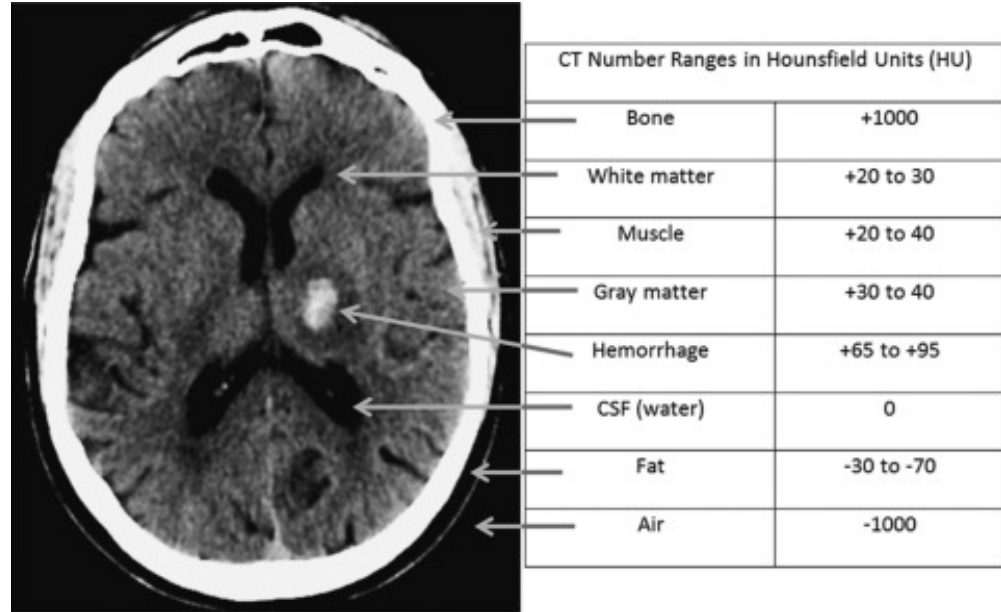
# Tomographic Imaging



# Hounsfield Units

The Hounsfield unit (HU) scale is a linear transformation of the original linear attenuation coefficient measurement into one in which the radiodensity of water is defined as 0 Hounsfield units (HU), while the radiodensity of air is defined as -1000 HU.

In a voxel with average linear attenuation coefficient  $\mu$ , the corresponding HU value is therefore given by:





# Houndsfield Units & Windowing

## Perception of images

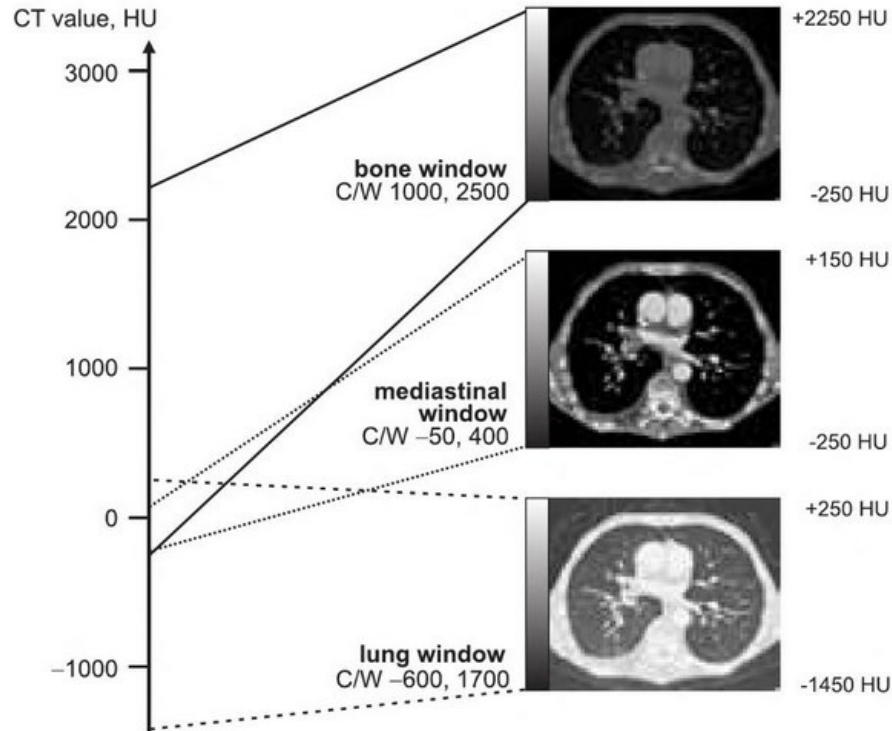
Grey scale – perception depth (human eye) – memory capacity – What is needed?

The human eye can approximately distinguish 80 shades of gray. Therefore, in photography a gray-scale-image is sufficiently represented by 8 bits, which allows 256 shades of gray. And  $g = 0$  represents black, and  $g = 255$  is white. This applies also for the most Computer Graphic-Cards.



<https://pixelcraft.photo.blog/tag/human-eye/>

# Hounsfield Units & Windowing



# Radiation Doses

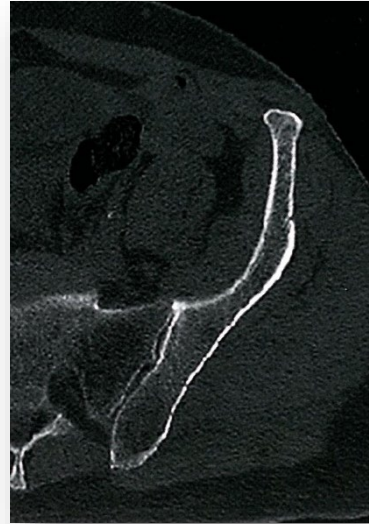
| Examination                   | Radiation dose (mSv) <sup>b</sup> | Time to accumulate comparable natural background dose |
|-------------------------------|-----------------------------------|-------------------------------------------------------|
| Computed tomography           |                                   |                                                       |
| Sinuses                       | 0.6                               | 2 mo                                                  |
| Head                          | 2.0                               | 8 mo                                                  |
| Chest                         | 7.0                               | 2 y                                                   |
| Chest (pulmonary embolism)    | 10.0                              | 3 y                                                   |
| Abdomen and pelvis            | 10.0                              | 3 y                                                   |
| Multiphase abdomen and pelvis | 31.0                              | 10 y                                                  |
| Radiography                   |                                   |                                                       |
| Extremity                     | 0.001                             | <1 d                                                  |
| Chest                         | 0.1                               | 10 d                                                  |
| Lumbar spine                  | 0.7                               | 3 mo                                                  |
| Abdomen                       | 1.2                               | 5 mo                                                  |
| Other                         |                                   |                                                       |
| Mammography                   | 0.7                               | 3 mo                                                  |
| Bone densitometry (DEXA)      | 0.001                             | <1 d                                                  |
| Nuclear medicine              |                                   |                                                       |
| Lung ventilation/perfusion    | 2.0                               | 8 mo                                                  |
| Bone scan                     | 4.2                               | 1 y, 4 mo                                             |
| Cardiac perfusion (sestamibi) | 12.5                              | 4 y                                                   |
| Fluoroscopy                   |                                   |                                                       |
| Barium swallow                | 1.5                               | 6 mo                                                  |
| Coronary angiography          | 5-15                              | 20 mo to 5 y                                          |

<sup>a</sup> DEXA = dual-energy x-ray absorptiometry.

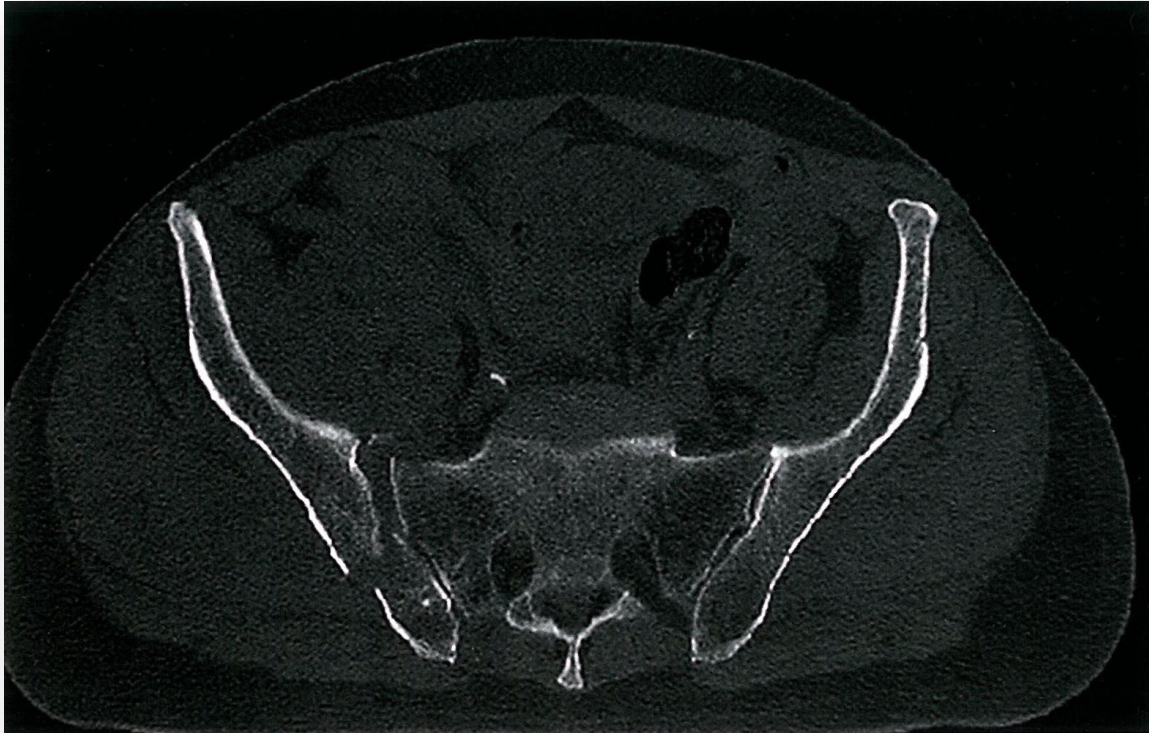
<sup>b</sup> These doses are effective doses, which are theoretical quantities proposed by the International Commission on Radiation Protection to assess the health risks of low doses of ionizing radiation.<sup>5</sup>

# Content

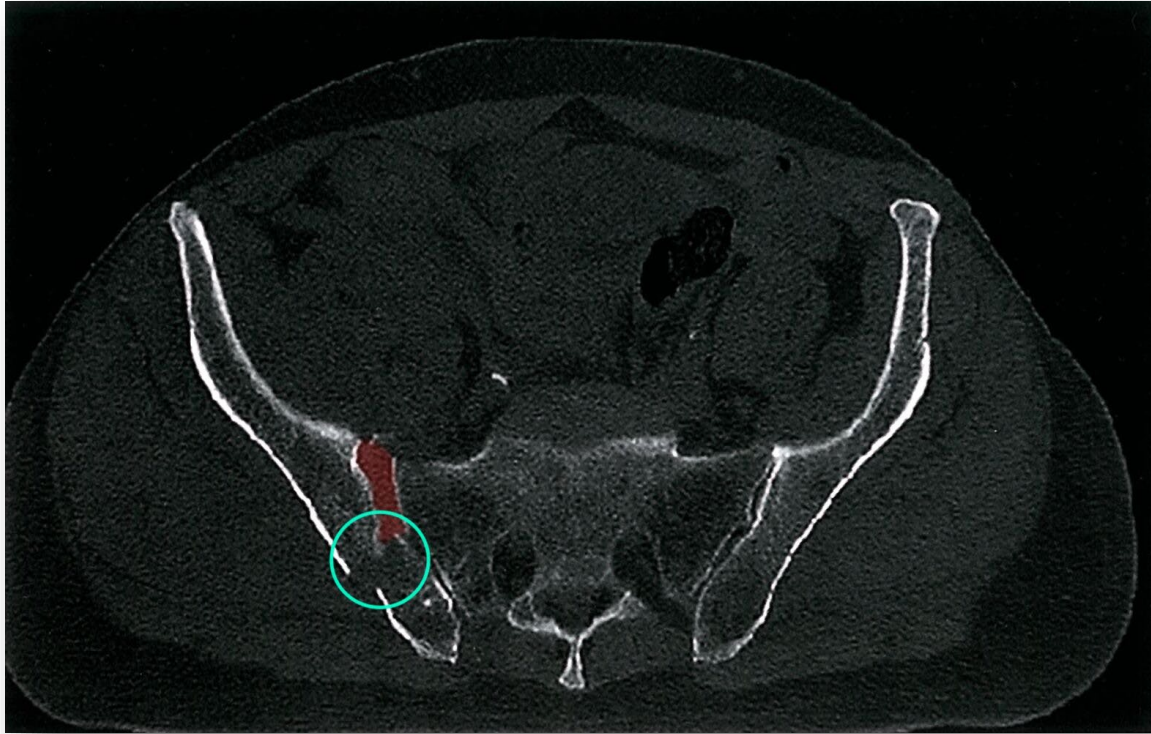
- History
- Principles
- **Applications**
- Data Aspects



# Bone CT

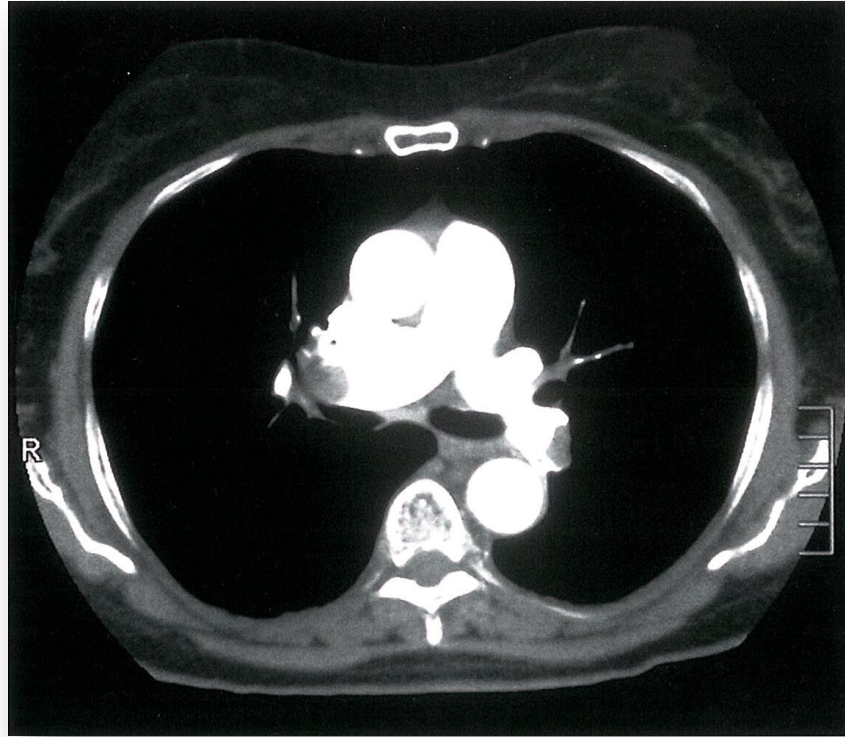


# Bone CT

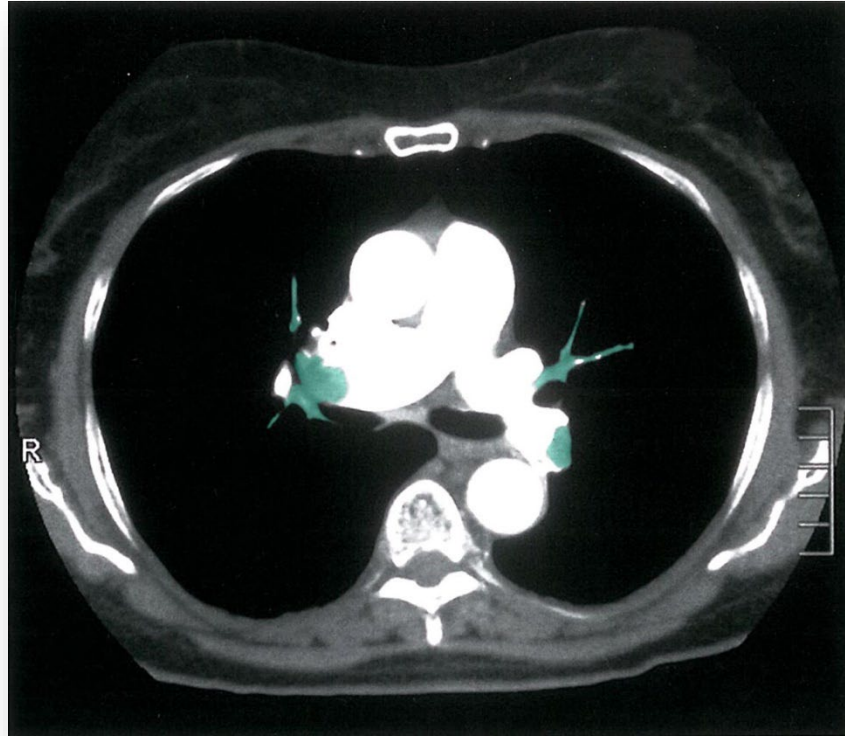




# Thorax CT



# Thorax CT





# Head CT



# Head CT



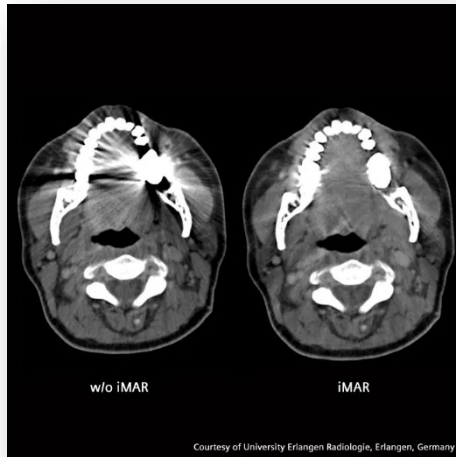
# Content

- History
- Principles
- Applications
- **Data Aspects**



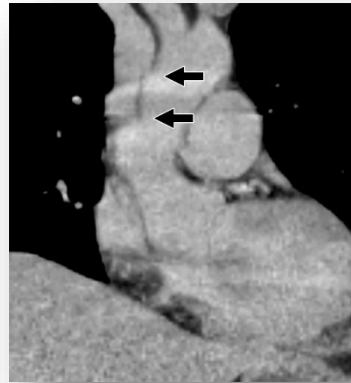
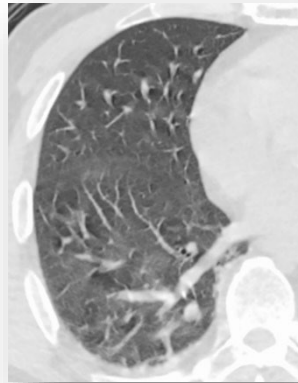
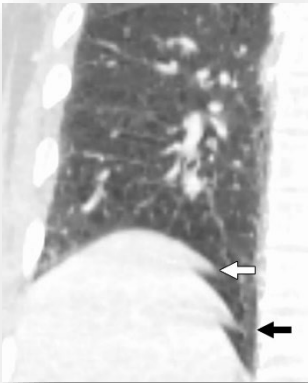
# Image Properties

- **Usage:** frequently used for numerous pathologies, because of availability, relative economy and despite (some) radiation
- **Projections:** typically 3D volume, viewed as 2D projections/slice images
- **Artifacts:** tomographic imaging can cause specific artifacts («streaks»)
  - Metal: high attenuation



# Image Properties

- **Usage:** frequently used for numerous pathologies, because of availability, relative economy and despite (some) radiation
- **Projections:** typically 3D volume, viewed as 2D projections/slice images
- **Artifacts:** tomographic imaging can cause specific artifacts («streaks»)
  - Metal: high attenuation
  - Movement: breathing, heart-beat, patient bulk motion



# Image Properties

- **Usage:** frequently used for numerous pathologies, because of availability, relative economy and despite (some) radiation
- **Projections:** typically 3D volume, viewed as 2D projections/slice images
- **Artifacts:** tomographic imaging can cause specific artifacts («streaks»)
  - Metal: high attenuation
  - Movement: breathing, heart-beat, patient bulk motion
  - Others: beam hardening, beam geometry-specific, scatter radiation etc.
- **Contrast:**
  - High contrast for bone and calcified structures
  - Low-intermediate soft tissue contrast
- **Noise:** Visible (Poisson-distributed) noise, especially in soft tissue
- **Spatial resolution:** typically high (~1-5 mm in all directions)

